

A trace-vanishing dichotomy for the Iwahori–Hecke algebra of the symmetric group

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Abstract

For partitions $\lambda = (\hat{\lambda}, 1^q)$ with $\ell(\hat{\lambda}) \geq 2$ and every row of $\hat{\lambda}$ of length ≥ 2 , we exhibit a specific element $\Omega^{(\lambda)}$ of the Iwahori–Hecke algebra $H_q(S_n)$ whose trace on the Specht module V^λ obeys a clean combinatorial dichotomy: $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0$ as a rational function of q if and only if a combinatorial threshold $\tau(\hat{\lambda}) := \max(0, q + 2\ell(\hat{\lambda}) - 1 - |\hat{\lambda}|)$ is at least 1. The proof combines four ingredients: (i) a *multi-row sign-kill meta-theorem* (Pillar 1) placing the image of $\Omega^{(\lambda)}$ in $\bigcap_j E_j^-$; (ii) a *right-absorption identity* $\Omega^{(\lambda)}(T_1 + 1) = (q + 1)\Omega^{(\lambda)}$; (iii) *per-SYT positivity* of every diagonal entry of $\Omega^{(\lambda)}$ in the asymmetric Hoefsmit basis; and (iv) an explicit *two-band SYT construction* T_{uniform} exhibiting a non-vanishing diagonal entry when $\tau(\hat{\lambda}) = 0$. The dichotomy is verified for all 5957 shapes with $|\lambda| \leq 25$. We close with a conjectural categorical lift in the homotopy category of Soergel bimodules.

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1 Introduction

When does the trace of a non-central element of the Iwahori–Hecke algebra vanish identically as a function of q ? For most algebra elements this question is meaningless or trivial. For a specific family $\{\Omega^{(\lambda)}\}$ indexed by partitions λ with at least two rows and a column-1 tail, we exhibit a clean combinatorial dichotomy.

The element $\Omega^{(\lambda)}$ is built out of products of $S_j := T_j + 1$. For $\nu \vdash N$ with $\ell = \ell(\nu)$, set $R'_k := S_{k-1}S_{k-2} \cdots S_1$ and

$$\Omega^{(\nu)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_N.$$

For $\widehat{\lambda}$ with $\ell(\widehat{\lambda}) \geq 2$ rows, all of length ≥ 2 , and $\lambda = (\widehat{\lambda}, 1^q)$, the threshold

$$\tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|)$$

encodes how much the tail length q exceeds the minimum compatible with the shape of $\widehat{\lambda}$.

Theorem 1.1 (Main dichotomy). *Let $\widehat{\lambda}$ be a partition with $\ell(\widehat{\lambda}) \geq 2$ rows, every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q) \vdash n$ for some $q \geq 1$. Then, as an identity of rational functions of q ,*

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0 \iff \tau(\widehat{\lambda}) \geq 1.$$

Equivalently, the trace vanishes identically iff $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$. The dichotomy has been verified on all 5957 shapes $|\lambda| \leq 25$ satisfying the row-length hypothesis.

Structure of the paper. Section 2 fixes notation: the Iwahori–Hecke algebra $H_q(S_n)$, the asymmetric Hoefsmit basis with $b' = 1$, and the descent eigenspaces E_j^+, E_j^- . Section 3 defines $\Omega^{(\lambda)}$ and τ , and tabulates first examples. Section 4 proves the forward direction ($\tau \geq 1 \Rightarrow \text{tr} = 0$) from Pillar 1 and right-absorption. Sections 5 and 6 prove the converse via per-SYT positivity and an explicit construction T_{uniform} . Section 7 proves strong sharpness $B^{(\lambda)} \cap E_{\tau+1}^- = \{0\}$, pinning the descent depth at exactly τ . Section 8 collects three independent per-SYT diagnostics for τ . Section 9 records the scope of computational verification. Section 10 states a conjectural categorical lift and the Demazure–Lusztig bridge to K -theory of the flag variety.

Related work. The seminormal basis of $H_q(S_n)$ goes back to Hoefsmit [1] and Murphy [2]; the cellular structure is due to Dipper–James [3]. Ram [4, 5] developed Markov traces and Jucys–Murphy product machinery in this setting. Tolmachov–Zhylynskiy [6] compute traces of products of polynomials in Jucys–Murphy elements in closed form; their formula motivates but does not directly settle the dichotomy because $\Omega^{(\lambda)}$ is not a central JM product (see Remark 9.1). The categorical home of our dichotomy is the derived horizontal trace $\text{HH}_*(\text{SBim}_n)$ studied by Gorsky–Hogancamp–Wedrich [7]; the relevant Specht-isotypic projectors P_λ are Hogancamp’s categorified Young symmetrizers [8, 9]. Recent work of Trinh–Williams [10] on Hecke trace formulas via partial Springer resolutions is, to our knowledge, the only published reference citing Tolmachov–Zhylynskiy.

Two routes to per-SYT positivity were investigated and *deflated*. The first, Goertzen–Williamson’s variational characterization of the seminormal basis [13], gives the right structural picture at $v = 1$ but does not transfer to generic q . The second, Bai–Gu–Guo–Liu’s 1423-avoidance criterion for non-negativity in certain pipe-dream rings [14], matches our diagonals on small shapes but is a different ring; the apparent bridge does not survive scrutiny. We comment on both in Section 10.

2 Preliminaries

2.1 The Iwahori–Hecke algebra

Let $H_q(S_n)$ denote the Iwahori–Hecke algebra of the symmetric group S_n over the field $\mathbb{Q}(q)$, with generators $T_1, \dots, T_{n-1}$ and relations

$$T_i^2 = (q - 1)T_i + q, \quad T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1}, \quad T_i T_j = T_j T_i \text{ for } |i - j| \geq 2.$$

The eigenvalues of each T_j on any finite-dimensional module are q and -1 . Set

$$S_j := T_j + 1, \quad j = 1, \dots, n - 1.$$

Lemma 2.1 (Square identity). $S_j^2 = (q + 1)S_j$ in $H_q(S_n)$ for every j .

Proof. $(T_j + 1)^2 = T_j^2 + 2T_j + 1 = ((q - 1)T_j + q) + 2T_j + 1 = (q + 1)T_j + (q + 1) = (q + 1)(T_j + 1)$. $\square$

Equivalently, $\frac{1}{q+1}S_j$ is an idempotent (the projector onto the $+q$ -eigenspace of T_j).

2.2 The Hoefsmit asymmetric seminormal basis

For $\nu \vdash N$, the Specht module V^ν has the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\nu)\}$ in which each T_j acts by an explicit two-dimensional block on each pair $\{T, s_j T\}$. We use the *asymmetric* normalization with $b' = 1$. Recall the content $c(r, k) := k - r$ of cell (r, k) and the axial distance $d_j(T) := c_{j+1}(T) - c_j(T)$ from j to $j + 1$. The action of T_j on v_T is

- (a) $d_j(T) = 1$ (same row): $T_j v_T = q v_T$.
- (b) $d_j(T) = -1$ (same column): $T_j v_T = -v_T$.
- (c) $|d_j(T)| \geq 2$ (so $s_j T \in \text{SYT}(\nu)$):

$$\begin{aligned} T_j v_T &= a_d v_T + v_{s_j T}, \\ T_j v_{s_j T} &= (a_d a_{-d} + q) v_T + a_{-d} v_{s_j T}, \end{aligned}$$

where $d = d_j(T)$ and $a_d := (q-1)q^d/(q^d-1)$.

2.3 Descent eigenspaces

For each $j \in \{1, \dots, N-1\}$, the operator T_j has eigenvalues $\{q, -1\}$ on V^ν , giving the orthogonal decomposition

$$V^\nu = E_j^+(V^\nu) \oplus E_j^-(V^\nu),$$

where

$$E_j^+ := \ker(T_j - q)|_{V^\nu}, \quad E_j^- := \ker(T_j + 1)|_{V^\nu}.$$

Equivalently, S_j acts on E_j^+ by the scalar $q+1$ and on E_j^- by 0. In the Hoefsmit basis, T_1 is diagonal: $T_1 v_T = q v_T$ if T has entries 1, 2 in the same row (*descent-0 at 1*), and $T_1 v_T = -v_T$ if in the same column (*descent-1 at 1*).

Remark 2.2 (Notational overload). The symbol q plays two distinct roles in this paper. Throughout, q is both the formal Hecke parameter (so that H_q is defined over $\mathbb{Q}(q)$) and the tail length of $\lambda = (\widehat{\lambda}, 1^q)$. This coincidence reflects the framework in which the four source papers were developed and we keep it; context disambiguates. The principal results, like Theorem 1.1, are about the trace as an identity of rational functions in q at the specific integer value q chosen as the tail length.

3 The element $\Omega^{(\lambda)}$ and the threshold τ

3.1 Definition of $\Omega^{(\nu)}$

For $\nu \vdash N$ with $\ell = \ell(\nu)$, set

$$R'_k := S_{k-1} S_{k-2} \cdots S_1 = (T_{k-1} + 1)(T_{k-2} + 1) \cdots (T_1 + 1), \quad R'_1 := 1,$$

and define

$$\Omega^{(\nu)} := R'_{\ell+1}^{(\nu)} R'_{\ell+2}^{(\nu)} \cdots R'_N^{(\nu)} \in H_q(S_n).$$

We will write $\Omega = \Omega^{(\lambda)}$ when λ is fixed.

3.2 The threshold τ

Throughout the remainder of the paper, $\widehat{\lambda}$ denotes a partition with $\ell(\widehat{\lambda}) \geq 2$ rows, every row of length ≥ 2 , and we set

$$\lambda := (\widehat{\lambda}_1, \dots, \widehat{\lambda}_{\ell(\widehat{\lambda})}, \underbrace{1, \dots, 1}_q) \vdash n, \quad n = q + |\widehat{\lambda}|,$$

with $\ell(\lambda) = \ell(\widehat{\lambda}) + q$. The threshold is

$$\tau(\widehat{\lambda}) := \max(0, q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|).$$

Equivalently, writing $|\widehat{\lambda}| - 2\ell(\widehat{\lambda}) = \sum_r (\widehat{\lambda}_r - 2)$ for the number of cells of $\widehat{\lambda}$ outside its first two columns,

$$\tau(\widehat{\lambda}) = \max(0, q - 1 - (|\widehat{\lambda}| - 2\ell(\widehat{\lambda}))).$$

The condition $\tau(\widehat{\lambda}) \geq 1$ is equivalent to $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$.

Example 3.1. We tabulate small cases.

$\widehat{\lambda}$	$\tau(\widehat{\lambda})$	smallest q with $\tau \geq 1$
(2, 2)	$\max(0, q - 1)$	$q = 2$
(3, 2)	$\max(0, q - 2)$	$q = 3$
(3, 3)	$\max(0, q - 3)$	$q = 4$
(2, 2, 2)	$\max(0, q - 1)$	$q = 2$
(3, 2, 2)	$\max(0, q - 2)$	$q = 3$

The smallest non-trivial dichotomy instance is $\widehat{\lambda} = (2, 2)$ between $q = 1$ ($\tau = 0$, $\text{tr} \neq 0$) and $q = 2$ ($\tau = 1$, $\text{tr} = 0$).

3.3 Statement of the main theorem

We restate Theorem 1.1 for reference.

Theorem 3.2 (Main dichotomy, restated). *With hypotheses on $\widehat{\lambda}$ as above and $\lambda = (\widehat{\lambda}, 1^q)$, $q \geq 1$,*

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0 \text{ in } \mathbb{Q}(q) \iff \tau(\widehat{\lambda}) \geq 1.$$

The forward direction is proved in Section 4; the converse in Sections 5–6.

4 Forward direction: structural nilpotence

The forward direction follows from two ingredients: a structural containment of the image ΩV^λ inside E_1^- (Pillar 1), and an algebraic identity in the ambient Hecke algebra (right-absorption). Together they show that, in the descent-at-1 block decomposition of V^λ , the matrix of $\Omega^{(\lambda)}$ is strictly lower triangular — in particular nilpotent of order at most 2, with vanishing trace.

4.1 Pillar 1: the multi-row meta-theorem

Theorem 4.1 (Pillar 1, multi-row meta-theorem). *Let $\widehat{\lambda}$ have $\ell(\widehat{\lambda}) \geq 2$ rows, every row of length ≥ 2 , and let $\lambda = (\widehat{\lambda}, 1^q)$ with $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$ (equivalently $\tau(\widehat{\lambda}) \geq 1$). Then*

$$\Omega^{(\lambda)} V^\lambda \subseteq \bigcap_{j=1}^{\tau(\widehat{\lambda})} E_j^-|_{V^\lambda}.$$

Proof sketch. Strong induction on $|\widehat{\lambda}|$. The two key ingredients are: (i) a *threshold-matching identity* (multi-row form) showing that every length-preserving contributor $\mu_{k,*}$ obtained from the E_{n-1}^+ decomposition preserves $\tau_0(\widehat{\lambda}) := q + 2\ell(\widehat{\lambda}) - 1 - |\widehat{\lambda}|$ exactly, in both the *regular* ($\widehat{\lambda}_k \geq 3$) and *shrinking* ($\widehat{\lambda}_k = 2$) cases; (ii) a *support rule*: if every T in the support of a subspace has $p(T) \geq P$ (where $p(T)$ is the column-1 identity-prefix length), then the subspace lies in $\bigcap_{j \leq P-1} E_j^-$. Vanisher pieces (corner-pair and same-row) inflate τ_0 by 2 and are killed by the IH plus an outer R'_ℓ factor whose rightmost S_1 annihilates the IH-provided E_1^- piece. Base case $\widehat{\lambda} = (2, 2)$ at $q \geq 2$ matches the May-14 two-row meta-theorem. See [15, Theorem 1] for the complete proof. $\square$

4.2 Right-absorption

Lemma 4.2 (Right-absorption). *As an element of $H_q(S_n)$,*

$$\Omega^{(\lambda)}(T_1 + 1) = (q + 1) \Omega^{(\lambda)}.$$

Proof. By Lemma 2.1, $(T_1 + 1)^2 = (q + 1)(T_1 + 1)$. The rightmost factor of $\Omega^{(\lambda)}$ is $R'_n = (T_{n-1} + 1) \cdots (T_1 + 1)$, ending in $(T_1 + 1)$. Therefore

$$R'_n \cdot (T_1 + 1) = (T_{n-1} + 1) \cdots (T_2 + 1) \cdot (T_1 + 1)^2 = (q + 1) R'_n.$$

Multiplying on the left by $R'_{\ell+1} \cdots R'_{n-1}$ gives the claim. $\square$

Remark 4.3. Lemma 4.2 is an identity in $H_q(S_n)$, valid on every H_q -module. In particular, for any $v \in E_1^-$ we have $(T_1 + 1)v = 0$, hence

$$(q + 1) \Omega v = \Omega(T_1 + 1)v = 0,$$

so $\Omega v = 0$ since $q + 1$ is a unit in $\mathbb{Q}(q)$. Thus $\Omega^{(\lambda)}$ *always* kills E_1^- regardless of the value of $\tau(\hat{\lambda})$.

4.3 The trace-vanishing corollary

Corollary 4.4 (Forward direction). *Under the hypotheses of Theorem 4.1 (in particular $\tau(\hat{\lambda}) \geq 1$),*

$$\Omega^{(\lambda)^2} = 0 \quad \text{on } V^\lambda, \quad \text{and} \quad \text{tr}_{V^\lambda}(\Omega^{(\lambda)}) = 0.$$

Proof. Order the Hoefsmit basis of V^λ by descent-at-1: place descent-0 tableaux first, descent-1 tableaux second. Write the matrix $M(\Omega)$ of $\Omega^{(\lambda)}$ in this ordering as the block matrix

$$M(\Omega) = \begin{pmatrix} A & B \\ X & D \end{pmatrix},$$

where rows/columns are split as (E_1^+, E_1^-) . We claim $A = 0$, $B = 0$, $D = 0$.

- For T descent-0 (i.e. $v_T \in E_1^+$): by Theorem 4.1, $\Omega v_T \in E_1^-$. Hence the column of $M(\Omega)$ at T has zeros in E_1^+ rows, so $A = 0$.
- For T descent-1 (i.e. $v_T \in E_1^-$): by Remark 4.3, $\Omega v_T = 0$. So columns B and D are both zero.

We conclude

$$M(\Omega) = \begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix},$$

and immediately $M(\Omega)^2 = 0$ and $\text{tr} M(\Omega) = 0$. $\square$

This proves the forward direction of Theorem 1.1. Note that we obtained the stronger conclusion $\Omega^2 = 0$, not merely $\text{tr}(\Omega) = 0$: the trace-vanishing dichotomy is in fact a *nilpotence* dichotomy.

5 Converse: per-SYT positivity

The converse direction requires showing that $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \neq 0$ whenever $\tau(\hat{\lambda}) = 0$. We work in the asymmetric Hoefsmit basis with $b' = 1$ and prove that every diagonal matrix entry of $\Omega^{(\lambda)}$ lies in the non-negative cone $\mathbb{Q}_{\geq 0}(q) := \{N(q)/D(q) : N, D \in \mathbb{Z}_{\geq 0}[q]\}$. A sum of elements of this cone vanishes iff each summand does, reducing the converse to exhibiting a single non-vanishing diagonal.

5.1 Hoefsmit positivity

Lemma 5.1 (Hoefsmit positivity). *Every entry of $S_i = T_i + 1$ on V^ν in the asymmetric Hoefsmit basis ($b' = 1$) lies in $\mathbb{Q}_{\geq 0}(q)$. Explicitly, with $d = d_i(T)$:*

- (i) *Diagonal entries: $[S_i]_{T,T} = q + 1$ if $d = 1$; 0 if $d = -1$; $[d + 1]_q/[d]_q$ if $|d| \geq 2$.*
- (ii) *Off-diagonal entries (when $|d| \geq 2$): $[S_i]_{s_i T, T} = 1$ and $[S_i]_{T, s_i T} = q[d - 1]_q[d + 1]_q/[d]_q^2$. All other off-diagonals vanish.*

Proof sketch. Direct computation from the Hoefsmit action. For $|d| \geq 2$, $[S_i]_{T,T} = a_d + 1 = q^{d+1} - 1 / (q^d - 1) = [d + 1]_q/[d]_q$; for $d \leq -2$, rewrite via $[-m]_q = -q^{-m}[m]_q$ to obtain $q[k - 1]_q/[k]_q \in \mathbb{Q}_{\geq 0}(q)$. The off-diagonal computation likewise yields $q[d - 1]_q[d + 1]_q/[d]_q^2$, manifestly in $\mathbb{Q}_{\geq 0}(q)$. See [17, Lemma 1]. $\square$

Corollary 5.2 (Product positivity). $\mathbb{Q}_{\geq 0}(q)$ is closed under sums and products. Hence every entry of $\Omega^{(\lambda)} = \prod_k R'_k$ in the asymmetric Hoefsmit basis is in $\mathbb{Q}_{\geq 0}(q)$.

Theorem 5.3 (Per-SYT positivity). *For every $T \in \text{SYT}(\lambda)$, the diagonal coefficient*

$$p_T(q) := [v_T](\Omega^{(\lambda)} v_T)$$

lies in $\mathbb{Q}_{\geq 0}(q)$. Consequently,

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) = \sum_{T \in \text{SYT}(\lambda)} p_T(q) \in \mathbb{Q}_{\geq 0}(q).$$

Proof. Immediate from Corollary 5.2: each p_T is a diagonal entry of $\Omega^{(\lambda)}$. $\square$

5.2 Reduction to one non-vanishing diagonal

Corollary 5.4 (Trace vanishes iff every diagonal vanishes). $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0$ as a rational function of q iff $p_T(q) \equiv 0$ for every $T \in \text{SYT}(\lambda)$.

Proof. The cone $\mathbb{Q}_{\geq 0}(q)$ consists of rational functions taking non-negative values at every real $q_0 > 0$ that is not a pole. A finite sum of such functions vanishing identically forces each summand to vanish at every common point of positivity, hence everywhere. $\square$

To complete the converse, we need a single $T \in \text{SYT}(\lambda)$ with $p_T(q) \neq 0$ when $\tau(\hat{\lambda}) = 0$.

5.3 Descent-1 kill and the stay-walk lower bound

Lemma 5.5 (Descent-1 kill). *If T is descent-1 at position 1, then $\Omega^{(\lambda)} v_T = 0$ and hence $p_T(q) = 0$.*

Proof. The rightmost factor of $\Omega^{(\lambda)}$ is $S_1 = T_1 + 1$, and $S_1 v_T = 0$ for T descent-1 at 1 by Lemma 5.1(i) with $d = -1$. $\square$

So only descent-0-at-1 SYTs can contribute non-trivially. Define an SYT T to be *column-descent-free* if no j has j and $j + 1$ in the same column at adjacent rows (Definition 6.1 below).

Lemma 5.6 (Stay-walk lower bound). *If $T \in \text{SYT}(\lambda)$ is descent-0 at 1 and column-descent-free, then*

$$p_T(q) \geq \prod_j [S_j]_{T,T}^{e_j} > 0 \quad \text{in } \mathbb{Q}_{\geq 0}(q),$$

where e_j counts occurrences of S_j in $\Omega^{(\lambda)}$.

Proof sketch. Expand $\Omega^{(\lambda)}v_T$ as a sum over Hoefsmit walks $T = T_0 \rightarrow T_1 \rightarrow \cdots \rightarrow T_N = T$, with each step weighted by a matrix entry of some S_j . By Corollary 5.2 every walk contributes a non-negative summand. The *stay walk* $T_0 = \cdots = T_N = T$ contributes $\prod_k [S_{j_k}]_{T,T}$. Under the descent-0-at-1 and column-descent-free hypotheses, every diagonal factor $[S_{j_k}]_{T,T}$ is one of $\{q+1\} \cup \{[d+1]_q/[d]_q : |d| \geq 2\}$ — all strictly positive in $\mathbb{Q}_{\geq 0}(q)$. The total p_T is bounded below by this stay-walk contribution. $\square$

Combined with Lemma 5.5, the converse direction reduces to:

Proposition 5.7 (Reduction). *If there exists $T^* \in \text{SYT}(\lambda)$ with descent-0 at 1 and no column-descents, then $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \neq 0$.*

6 Converse: existence of T^*

We close the converse by giving an explicit construction T_{uniform} of a descent-0-at-1 column-descent-free SYT of shape λ , valid for every $\tau(\widehat{\lambda}) = 0$ shape. The construction is uniform across the regimes $q = 1$ and $q \geq 2$ (in contrast to the earlier case-split version that used the row-superstandard tableau for $q = 1$ and an interleaved construction for $q \geq 2$).

Definition 6.1. An SYT T of shape λ has a *column descent at j* if the entries j and $j+1$ sit in the same column at adjacent rows. T is *column-descent-free* if no j is a column descent.

6.1 The construction T_{uniform}

We use 0-indexed coordinates. Let $L = \ell(\widehat{\lambda})$, $M = |\widehat{\lambda}| - 2L$, $n = 2L + M + q$. Set

$$k := \#\{r \in \{0, \dots, L-1\} : \widehat{\lambda}_r \geq 3\}.$$

Column-0 values. For $r \in \{0, \dots, L+q-1\}$, define

$$v_r := \begin{cases} 2r+1, & 0 \leq r < k, \\ n-2(L+q-1-r), & k \leq r \leq L+q-1. \end{cases} \quad (1)$$

Place $T_{\text{uniform}}(r, 0) := v_r$. These form a band of small odd values $1, 3, \dots, 2k-1$ at the top of column 0 (the “long-row band”) and a band of large odd values $\dots, n-4, n-2, n$ at the bottom (the “tail band”).

Remaining cells. Let $w_1 < w_2 < \cdots < w_{L+M}$ be the remaining values of $\{1, \dots, n\} \setminus \{v_r\}$ in increasing order. Fill the subdiagram $\widehat{\lambda}' := (\widehat{\lambda}_0 - 1, \dots, \widehat{\lambda}_{L-1} - 1)$ row-by-row: row r of $\widehat{\lambda}'$ receives values $w_{a_r+1}, \dots, w_{a_r+\widehat{\lambda}_r-1}$ at columns $1, \dots, \widehat{\lambda}_r - 1$ of λ , where $a_r := \sum_{i < r} (\widehat{\lambda}_i - 1)$.

Example 6.2 ($\widehat{\lambda} = (3, 2)$ at $q = 1$). Here $L = 2$, $M = 1$, $k = 1$, $n = 6$, $\lambda = (3, 2, 1)$. Column-0 values: $v_0 = 1, v_1 = 4, v_2 = 6$. Remaining $\{2, 3, 5\}$ fill $\widehat{\lambda}' = (2, 1)$ row-superstandard: row 0 receives 2, 3; row 1 receives 5. Resulting tableau:

$$T_{\text{uniform}} = \begin{array}{|c|c|c|} \hline 1 & 2 & 3 \\ \hline 4 & 5 & \\ \hline 6 & & \\ \hline \end{array}$$

Rows and columns are strictly increasing; entries 1, 2 are in row 0 (descent-0 at 1); no $j, j+1$ pair shares a column.

6.2 Three elementary lemmas

The verification that T_{uniform} has the required properties reduces to three elementary inequalities about v_r and w_i .

Lemma 6.3 (Column-0 gaps). $v_0 = 1$, $v_{L+q-1} = n$, and $v_{r+1} - v_r \geq 2$ for all r .

Lemma 6.4 (Row condition $v_r < w_{a_r+1}$). For every $r \in \{0, \dots, L-1\}$, the column-0 value v_r is strictly less than the first remaining value w_{a_r+1} in row r .

Lemma 6.5 (No column descents). No two consecutive integers in T_{uniform} share a column at adjacent rows.

6.3 The main existence theorem

Theorem 6.6 (Existence of T^*). For every $\widehat{\lambda}$ with $\ell(\widehat{\lambda}) \geq 2$, each row of length ≥ 2 , and every $q \geq 1$ satisfying $\tau(\widehat{\lambda}) = 0$, the tableau T_{uniform} of shape $\lambda = (\widehat{\lambda}, 1^q)$ is a column-descent-free SYT with $T_{\text{uniform}}(0, 0) = 1$ and $T_{\text{uniform}}(0, 1) = 2$.

Proof. By Lemma 6.3, column 0 is strictly increasing. By Lemma 6.4, each row is strictly increasing at the column-0-to-column-1 transition; strict increase within $\widehat{\lambda}'$ follows from the increasing w -sequence and row-major filling. Column strictness within $\widehat{\lambda}'$ likewise. Hence T_{uniform} is a valid SYT. The values $T_{\text{uniform}}(0, 0) = v_0 = 1$ and $T_{\text{uniform}}(0, 1) = w_1 = 2$ are immediate. By Lemma 6.5, no column descents. $\square$

Corollary 6.7 (Converse direction). If $\tau(\widehat{\lambda}) = 0$, then $\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \neq 0$.

Proof. Combine Theorem 6.6 with Proposition 5.7. $\square$

Together with Corollary 4.4, this closes the proof of Theorem 1.1.

7 Sharpness at $\tau + 1$

Pillar 1 (Theorem 4.1) places ΩV^λ inside $\bigcap_{j=1}^\tau E_j^-$. A natural question is whether this depth is best possible. We prove that it is, in the strong form $B^{(\lambda)} \cap E_{\tau+1}^- = \{0\}$.

7.1 Statement

Write $B^{(\lambda)} := \Omega^{(\lambda)} V^\lambda$ for the chain image.

Theorem 7.1 (Strong sharpness). Under the standing hypothesis ($\ell(\widehat{\lambda}) \geq 2$, every row ≥ 2 , $q \geq |\widehat{\lambda}| - 2\ell(\widehat{\lambda}) + 2$),

$$B^{(\lambda)} \cap E_{\tau+1}^-|_{V^\lambda} = \{0\},$$

equivalently, $S_{\tau+1} := T_{\tau+1} + 1$ is injective on $B^{(\lambda)}$.

Corollary 7.2. $\sup\{j : B^{(\lambda)} \subseteq E_j^-|_{V^\lambda}\} = \tau(\widehat{\lambda})$ exactly.

Theorem 7.1 is proved in [19]; combined with Pillar 1 [15], it gives the exact descent depth in Corollary 7.2.

7.2 The one-fact proof

The proof rides on a single structural fact about the outer product $\mathcal{O} := \prod_{j=\ell}^{n-2} (T_j + 1)$ appearing in the Pillar 1 reduction theorem, together with a geometric commutativity observation.

Lemma 7.3 (Outer product injectivity). *$\mathcal{O}$ is injective on $E_{n-1}^+|_{V^\lambda}$, with image in $E_\ell^+|_{V^\lambda}$.*

The proof is a chain application of the adjacent-injectivity identity $E_j^- \cap E_{j+1}^+ = 0$: for $v \in E_{n-1}^+$, $(T_{n-2} + 1)v \in E_{n-2}^+$ and the restriction $(T_{n-2} + 1)|_{E_{n-1}^+}$ is injective; iterating down to $j = \ell$ gives the claim.

Lemma 7.4 (Commutativity). *Under the standing hypothesis, $S_{\tau+1}$ commutes with every factor of $\mathcal{O}$ and with each contributor Phi-embedding $\Phi_{k,*}$.*

Proof sketch. The key inequality is

$$\ell - (\tau + 1) = |\widehat{\lambda}| - \ell(\widehat{\lambda}) \geq \ell(\widehat{\lambda}) \geq 2,$$

using every-row- ≥ 2 and $\ell(\widehat{\lambda}) \geq 2$. Hence every factor index $j \geq \ell$ of $\mathcal{O}$ satisfies $|j - (\tau + 1)| \geq 2$, and Hecke generators at distance ≥ 2 commute. The Phi-embeddings are T_i -equivariant for $1 \leq i \leq n - 3$, and the same inequality places $\tau + 1$ in this range. $\square$

Proof of Theorem 7.1 (sketch). Strong induction on $|\widehat{\lambda}|$. Let $w \in B^{(\lambda)}$ with $S_{\tau+1}w = 0$. By the Pillar 1 reduction theorem, $w = \mathcal{O} \sum_k \Phi_{k,*}(u_k)$ uniquely with $u_k \in B^{(\mu_{k,*})}$. By Lemma 7.4, $S_{\tau+1}$ commutes with $\mathcal{O}$ and with each $\Phi_{k,*}$, so

$$0 = S_{\tau+1}w = \mathcal{O} \sum_k \Phi_{k,*}(S_{\tau+1}u_k).$$

The inner sum lies in $E_{n-1}^+|_{V^\lambda}$; by Lemma 7.3 each $\Phi_{k,*}(S_{\tau+1}u_k) = 0$, and disjoint Phi-supports plus injectivity of each $\Phi_{k,*}$ give $S_{\tau+1}u_k = 0$. By the threshold-matching identity, $\tau(\hat{\mu}_{k,*}) = \tau(\widehat{\lambda})$ for every contributor, so the IH at $\mu_{k,*}$ gives $u_k = 0$, hence $w = 0$. $\square$

7.3 Hook base case

The induction bottoms out on the hook layer $\widehat{\lambda} = (k_h)$ (length 1 shapes, formally outside the standing hypothesis but reached as contributor inner shapes). The hook analogue is

Lemma 7.5 (Hook sharpness). *For $\mu = (k_h, 1^{m_h})$ with $k_h \geq 2$ and $m_h \geq k_h$, $S_{m_h-k_h+2}$ is injective on $B^{(\mu)}$.*

Proof sketch. Induction on k_h with base case $k_h = 2$ by direct Hoefsmit computation: $B^{(2,1^{m_h})} = E_{m_h+1}^+|_{V_\mu}$ is a line, spanned by $w_0 = v_{T^+} + \beta v_{T^-}$ with T^+ placing the top two values at corners $(1, 2)$ and $(m_h + 1, 1)$ and T^- swapping them. In T^+ , the cells of m_h and $m_h + 1$ are non-adjacent (axial distance $m_h \neq \pm 1$), giving a non-degenerate T_{m_h} -block with nonzero off-diagonal; in T^- , same column, so $S_{m_h}v_{T^-} = 0$. The image $S_{m_h}w_0$ therefore has a nonzero coefficient on a basis vector outside $\text{supp}(B^{(\mu)})$, witnessing injectivity. The inductive step uses a hook analogue of the Pillar 1 reduction with $\mathcal{O}^{(\mu)} = \prod_{j=m_h+1}^{m_h+k_h-2} (T_j + 1)$ and contributor $\mu_{1,*} = (k_h - 1, 1^{m_h-1})$, the same commutativity argument as above, and the IH on k_h . $\square$

7.4 Verification

The full rank $\dim S_{\tau+1}(B^{(\lambda)}) = \dim B^{(\lambda)}$ has been verified on 14 datapoints (eight multi-row shapes and six hooks) computed via a sparse Hoefsmit implementation mod $P = 100003$ at $q = 11$; see [19, Table 1].

8 Per-SYT diagnostics

The converse direction (Section 5) was proved by showing each diagonal $p_T(q) := [v_T](\Omega v_T)$ lies in the positive cone $\mathbb{Q}_{\geq 0}(q)$ and exhibiting a non-vanishing T_{uniform} . In this section we collect three independent invariants that pin down which T 's contribute and which do not, framing the trace dichotomy as a confluence of structural diagnostics rather than a single identity.

8.1 Diagnostic 1: per-SYT positivity

Theorem 8.1 (Per-SYT positivity, restated). *For every $T \in \text{SYT}(\lambda)$, the diagonal $p_T(q) \in \mathbb{Q}_{\geq 0}(q)$ in lowest terms.*

This is Theorem 5.3, immediate from Hoefsmit positivity (Lemma 5.1) and closure of the cone under sums and products. Combined with the existence of T_{uniform} (Theorem 6.6), it closes the converse: $\tau(\hat{\lambda}) = 0$ forces $p_{T_{\text{uniform}}}(q) \neq 0$ on a stay-walk lower bound, so the trace is non-vanishing. The cleavage is structural — there is no fortuitous cancellation between T -summands, since none can cancel against another in a positive cone.

8.2 Diagnostic 2: leftmost-factor SC vanishing

Theorem 8.2 (Per-SYT leftmost-factor sign-kill). *Let $\lambda \vdash n$ with $\ell = \ell(\lambda)$. If the values ℓ and $\ell+1$ sit in the same column of T (pair $(\ell, \ell+1)$ is SC at T , equivalently $S_{\ell}v_T = 0$), then $p_T(q) = 0$.*

Two-line proof. Write $\Omega = R'_{\ell+1}R'_{\ell+2} \cdots R'_n = S_{\ell} \cdot R'_{\ell}R'_{\ell+2} \cdots R'_n =: S_{\ell} \cdot X$. Then $\Omega v_T = S_{\ell}(Xv_T) \in \text{im}(S_{\ell}) = E_{\ell}^+|_{V^{\lambda}}$, while $v_T \in E_{\ell}^-|_{V^{\lambda}}$ by SC. Project the seminormal expansion $\Omega v_T = \sum_{T'} c_{T,T'} v_{T'}$ onto E_{ℓ}^- along E_{ℓ}^+ : the left side is 0, the right side has $c_{T,T''} = 0$ for every $T'' \in \text{SYT}^{\text{SC}} \text{ at } \ell$; specialise $T'' = T$. $\square$

Remark 8.3. The theorem holds without any restriction on λ (multi-row, τ , staircase). It is the per-SYT refinement of the universal containment $B^{(\lambda)} \subseteq E_{\ell}^+|_{V^{\lambda}}$ obtained from the same leftmost-factor identity. See [20] for the full exposition.

At $\lambda = (3, 3, 1, 1, 1)$, an empirical sweep produced 108 zero diagonals out of $|\text{SYT}(\lambda)| = 120$. Theorem 8.2 accounts for 26 of these — exactly the descent-0 SYTs with pair $(5, 6)$ same-column. In particular, the historically mysterious zero at

$$T_0 = ((1, 2, 5), (3, 4, 6), (7), (8), (9))$$

is explained by SC at $(5, 6)$ (value 5 at row 0 column 2, value 6 at row 1 column 2).

8.3 Diagnostic 3: Pillar-1-side SR vanishing

The Pillar-1 containment $B^{(\lambda)} \subseteq E_j^-$ for $j = 1, \dots, \tau$ has a per-SYT shadow obtained by the same projection argument run at index $j \leq \tau$:

Corollary 8.4 (Pillar-1-side per-SYT vanishing). *If $T \in \text{SYT}(\lambda)$ has pair $(j, j+1)$ same-row for some $j \in \{1, \dots, \tau\}$ (i.e., $v_T \in E_j^+$), then $p_T(q) = 0$.*

Theorems 8.2 and Corollary 8.4 provide *three* independent invariants pinning τ : structural depth (Pillar 1 + sharpness), trace vanishing (Theorem 1.1), and per-SYT leftmost/rightmost diagnostics. Any future categorical lift (Conjecture 10.1) must unify all three.

8.4 Open: companion diagnostic and interior zeros

SR companion conjecture. Theorem 8.2 fires at the ℓ -side via SC. A mirror statement is expected at the τ -side: *if pair $(j, j+1)$ is same-row at T for some $j \in \{1, \dots, \tau\}$, then $p_T(q) = 0$.* This is exactly Corollary 8.4 above, so the “mirror” is in fact already proved. A finer companion conjecture concerns the rightmost-factor identity $\Omega = Y \cdot S_1$ extended to inner positions: a one-page proof is queued in the agent’s proof-attempts file.

Interior D-class zeros. The $108 - 26 = 82$ zero diagonals at $\lambda = (3, 3, 1, 1, 1)$ not explained by Theorem 8.2 split further: 14 are descent-1 trivial vanishings (pair $(1, 2)$ SC, $\Omega v_T = 0$ outright), and 68 are descent-0 SYTs with pair $(\ell, \ell+1)$ diagonal (D-class). For these, v_T has nonzero components in both E_ℓ^+ and E_ℓ^- , and the seminormal projection no longer forces $c_{T,T} = 0$ directly. These “interior cancellations” are not explained by any of the three diagnostics above; characterising them is the natural next-horizon refinement of per-SYT analysis.

9 Computational verification

The four ingredients have been verified independently.

Scope. The dichotomy was tested on every shape $\lambda = (\hat{\lambda}, 1^q)$ with $|\lambda| \leq 25$ satisfying the hypothesis $\ell(\hat{\lambda}) \geq 2$ and every row ≥ 2 , yielding 5957 shapes in total: 210 with $|\lambda| \leq 14$; 832 with $|\lambda| \leq 18$; 2688 with $|\lambda| \leq 22$; 5957 with $|\lambda| \leq 25$.

Methodology.

- Symbolic computation in `sympy` verified Hoefsmit positivity (Lemma 5.1) for axial distances $|d| \leq 6$.
- Direct construction confirmed the T_{uniform} algorithm produces column-descent-free SYTs on all 5957 shapes (`t_unified.py`).
- Pillar 1 and trace-vanishing were verified via sparse application of $\Omega^{(\lambda)}$ to random vectors over $\mathbb{F}_p$ with $p = 100003$, $q = 11$, with rank stability checks across multiple probes.

Refuted alternatives. Three structurally different routes were investigated and ruled out by computational data:

- A q -binomial product ansatz for $\text{tr}(\Omega)$ failed at $\lambda = (4, 3, 1^2)$.
- The non-crossing matching factorisation (motivated by Jucys–Murphy products) gave rank 12 where Catalan-counting predicted 5.
- The Goertzen–Williamson variational characterisation [13] of the seminormal basis is only sharp at $v = 1$ (Soergel calculus specialisation), giving no generic- q positivity.

- The Bai–Gu–Guo–Liu 1423-avoidance criterion [14] matches our diagonals on 7/7 tested shapes, but the underlying rings differ (theirs is $\mathbb{Z}_{\geq 0}[\beta]$ pipe-dream counts; ours is $\mathbb{Q}_{\geq 0}(q)$ with cyclotomic poles).

Remark 9.1. Ram’s content multiset formula [5] and the Tolmachov–Zhylynskyi closed form for $\text{tr}(\prod p_i(J_i))$ [6] suggest a route via Jucys–Murphy products. However, the multiset $\{c_i(T) : i\}$ is a *shape* invariant, so symmetric JM expressions cannot detect a $\hat{\lambda}$ -vs- q dichotomy. Trace-vanishing of $\Omega^{(\lambda)}$ requires a position-asymmetric / non-central JM expression; the right-absorption proof bypasses this entirely.

10 Outlook: a categorical lift

The forward direction of Theorem 1.1 proved strictly more than trace vanishing: it gave $\Omega^{(\lambda)2} = 0$ on V^λ , with the matrix of $\Omega^{(\lambda)}$ in the descent-at-1 block decomposition having the form $\begin{pmatrix} 0 & 0 \\ X & 0 \end{pmatrix}$. This structural rigidity invites a categorification.

10.1 The conjecture

Conjecture 10.1 (Categorical dichotomy). *Let $\tilde{\Omega}^{(\lambda)}$ be a Rouquier complex categorifying $\Omega^{(\lambda)}$ as an object of the homotopy category of Soergel bimodules $K^b(\text{SBim}_n)$, and let P_λ denote Hogancamp’s categorified Young symmetrizer for λ . Then*

$$\text{tr}_{V^\lambda}(\Omega^{(\lambda)}) \equiv 0 \iff \text{HH}_*(P_\lambda \cdot \tilde{\Omega}^{(\lambda)} \cdot P_\lambda) = 0$$

in bidegree ($q\text{-deg} = |\lambda|$, $\text{hom-deg} = 0$).

The Specht-isotypic projector P_λ realises V^λ as the λ -component of the universal Markov trace; the conjecture says that the $\hat{\lambda}/\tau$ dichotomy survives intact at the categorical level.

10.2 What the lift should achieve

Specht-isotypic decomposition. The honest gap is that the Specht-isotypic decomposition

$$\text{Tr}(\text{SBim}_n) = \bigoplus_{\lambda} \text{Tr}^\lambda$$

of the horizontal trace into derived blocks is not yet explicitly written down in the derived setting [7]. Hogancamp’s categorified Young symmetrizers [8, 9] provide P_λ , and the conjectural decomposition should be $\text{Tr}^\lambda = \text{Tr}(\text{SBim}_n \cdot P_\lambda)$.

Lifting right-absorption. The identity $S_1^2 = (q+1)S_1$ should categorify to an explicit homotopy / contractibility statement in the categorified bar complex. This is the categorical analogue of “ Ω ends in S_1 , and $S_1^2 = (q+1)S_1$ ” — the load-bearing identity behind Lemma 4.2.

10.3 Reading list

Reading list, in suggested order, for tackling Conjecture 10.1:

- (1) [11] ([arXiv:2507.10061](https://arxiv.org/abs/2507.10061)) — Idempotents, traces, and dimensions in Hecke categories. Sits at the idempotent / trace junction; may already contain a Specht-isotypic decomposition.

- (2) Trinh–Williams [10] ([arXiv:2601.17293](#)) — Partial Springer resolutions and Hecke trace formulas.
- (3) Hogancamp [8] ([arXiv:1505.08148](#)) and Abel–Hogancamp [9] ([arXiv:1510.05330](#)) — Categorified Young symmetrizers.
- (4) Gorsky–Hogancamp–Wedrich [7] ([arXiv:2002.06110](#)) — Derived horizontal trace of Soergel bimodules.
- (5) Wedrich’s 2025 survey [12] ([arXiv:2509.08478](#)) for context on the link-homology / TQFT environment.

10.4 The Demazure–Lusztig bridge

A second next-horizon direction is the decategorified bridge between the LR / Schubert structure constants and the Markov-trace evaluation, mediated by the Demazure–Lusztig operator on $K_T(G/B)$. Five published papers [21, 22, 23, 24, 25] use this operator to evaluate equivariant K -theory pairings on the flag variety in terms of Hecke-algebra characters. The trace $\mathrm{tr}_{V^\lambda}(\Omega^{(\lambda)})$ should equal a localisation residue on the λ -fixed point of $K_T(G/B)$ via Atiyah–Bott / AMSS, and $\tau(\hat{\lambda}) \geq 1$ should manifest as residue cancellation in this localisation expansion. This is the natural next-horizon for a *structural* (as opposed to algebraic) understanding of τ : the threshold appears here as the height of a torus-fixed point inside a Schubert stratification, not as a combinatorial count. Working out exactly which Demazure–Lusztig word evaluates to $\Omega^{(\lambda)}$ and extracting the residue cancellation conjecturally encoded by $\tau \geq 1$ is left for future work.

10.5 What this paper does not address

We do not address the non-stable regime $q < |\hat{\lambda}| - 2\ell(\hat{\lambda}) + 2$ where the standing hypothesis fails, or refinement of Theorem 5.3 to $p_T(q) \in \mathbb{Z}_{\geq 0}[q]$ (the trace is empirically a non-negative-coefficient polynomial, but our Hoefsmit-based proof gives only rational positivity); see Remark 3 of [17]. The interior D-class zero diagonals at $\lambda = (3, 3, 1, 1, 1)$ (Section 8.4) remain unexplained by the three diagnostics collected here.

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